Nenavath Ramu Naik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.63
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education (TSBIE))	2023	93.50%
Narayana e-Techno School	Class X (Board of Secondary Education, Telangana State)	2021	100.00%

POSITIONS OF RESPONSIBILITY

- **Milan - Multimedia Coordinator:** Coordinated multimedia activities, event coverage, and participant engagement for IIT Hyderabad's annual techno-cultural-sports festival.
- **NSS Volunteer:** Participated in community service initiatives, promoting social awareness, teamwork, and civic responsibility.
- **Inter IIT Operations Coordinator:** Managed event logistics and coordinated with multiple teams to ensure smooth execution of Inter IIT activities.
- **Departmental Representative (DR):** Acted as a liaison between students and faculty, communicating academic concerns and facilitating timely resolutions.
- **CMC Coordinator (Canteen Monitoring Committee):** Monitored canteen operations, coordinated with students and staff, and addressed service-related concerns to ensure smooth functioning.
- **HR Coordinator:** Supported volunteer coordination, team communication, task allocation, and event operations to ensure effective team management.

SKILLS

Programming Languages: Java, C++, Python, Go, SQL

Data Structures & Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, System Design

Web Technologies: HTML, CSS, JavaScript, ReactJS, NodeJS, ExpressJS

Databases: PostgreSQL, MySQL, MongoDB

Machine Learning & NLP: NumPy, Pandas, SciPy, Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face, NLTK

AI & LLM Frameworks: LangChain, LangGraph

C++: C++ Threads

Relevant Coursework: Machine Learning (Andrew Ng, Coursera), Operating Systems, Databases, Data Structures & Algorithms, Introduction to Programming

WORK EXPERIENCE

Civil Engineering Intern -- BARC, Mumbai

- Structural analysis and design of a G+5 RCC building using SAP2000.

- Worked with structural modelling, load application, and design analysis.

PROJECTS

AI Chieftain -- Full-Stack AI Concierge | *React, Node.js, MongoDB, Socket.IO, Gemini API, Twilio, RAG*

- Built a full-stack **agentic AI concierge** integrating Gemini API with MongoDB for dynamic function calling, live bookings, and service management.
- Implemented **RAG-based personalization** using user history to generate context-aware and sentiment-aware responses.
- Developed a real-time **Socket.IO staff dashboard** and proactive notification system with voice-enabled web interactions and Twilio telephony.

Agentic Document Editing System | *LangChain, LangGraph, AutoGPT, AutoGen, RAG, Local LLMs*

- Developed an **agentic document-processing system** using LangGraph, prompt engineering, RAG, and local LLMs for long-document editing workflows.
- Evaluated AutoGPT and AutoGen agents and optimized the workflow for improved document-processing efficiency, accuracy, and scalability.

Chat Room | *C++, Boost.Asio, Multithreading*

- Developed a **multi-client real-time chat application** using asynchronous networking and multithreading to handle concurrent connections.
- Implemented session/room management, message encoding/decoding, and dynamic client tracking for reliable message delivery.
- Optimized the server using non-blocking communication to improve scalability and responsiveness.

Map Pathfinding Visualiser | *ReactJS, Overpass API, Graph Algorithms*

- Built an interactive React application to visualize *Dijkstra, A, Bidirectional Search, and Greedy Search** algorithms.
- Integrated Overpass API and OpenStreetMap data to retrieve and filter road networks and construct graphs for route visualization.
- Added animated pathfinding and playback-speed controls for interactive algorithm visualization.

Next-Generation Liquid Membranes for Manganese Recovery | *Membrane Technology, Metal Extraction, Environmental Engineering*

- Developed and optimized a liquid membrane system for selective **Mn^2 extraction from industrial wastewater**.
- Conducted PVC-THF membrane synthesis and evaluated extraction performance under varying feed concentrations and contact times.
- Achieved up to **81.24% manganese recovery** and identified key parameters affecting extraction efficiency.